Reg. No									

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore–607001
B.Sc Degree Examination - APRIL 2021

CH203T – INORGANIC CHEMISTRY - I

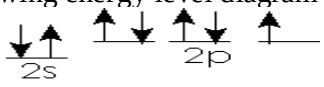
Time : 3 hrs **Max Marks :75**

SECTION – A (20X1=20)

Answer **ALL** Questions

I. Choose the correct answer

1. Which of the following has highest first ionization potential?
   a) Carbon b) Oxygen c) Nitrogen d) Boron
2. Mark the correct statement:
   a) Na^+ is smaller than Na atom b) Na^+ is larger than Na atom
   c) CI^- is smaller than CI atom d) CI^- and CI are equal in size
3. The alkali metals all have similar properties because
   a) They all have one electron in their outer shells b) They have similar atomic numbers
   c) They are all in the periodic table d) They are all equally reactive
4. If you heat metallic calcium in a test tube with a similar mass of sulfur, After a few seconds, a fireball ensued producing
   a) Calcium silicate b) Calcium sulfide c) Calcium sulfite d) Calcium sulfate
5. The following energy level diagram represents the outermost shell of which ground state element?



   a) N b) Cl c) F d) C e) Mg
6. Which one of the following is the correct bond angle between atoms adopting a trigonal planar geometry?
   a) 180º b) 109.5º c) 90º d) 120º
7. N has 5 valence electrons and it is diatomic. How many covalent bonds are there in a N_2 molecule?
   a) One b) Two c) Three d) Four
8. Hydrogen Fluoride has a (an) _____ bond and its chemical formula is _____.
   a) Covalent HF b) Ionic HF c) Covalent H_2F d) Ionic H_2F
9. Which of the following is wrong regarding the structure of Diborane ?
   a) 4 terminal hydrogen atoms and 2 bridged hydrogen atoms
   b) The terminal hydrogen atoms and boron atoms lie in the same plane
   c) 4 two centre bonds 2 three centre bonds
   d) Boron is sp^3 hybridised
10. Carbon monoxide has ten bonding electrons and four antibonding electrons. Therefore it has a bond order of
    a) 3 b) 7 c) 1 d) 5/2

II. Fill up the blanks

11. The energy required to promote an electron from a lower energy orbital to a higher energy orbital is _____
12. The bond formed when Ca combines with Cl_2 is _____.
13. The maximm number of unpaired electrons in d orbital is _____

14. When a F atom gains an electron, it gets a charge of _____

15. The distribution of electrons in various orbitals is known as_____.

III. Match the following

16. Hunds rule	- H_2O
17. Hydrogen bonding	- Electron deficient
18. BH_3	- Polarisation
19. Fajon's rule	- Sidgwick's powell
20. Molecular shapes	- Maximum multiplicity

SECTION – B (10X2=20)

Answer any **TEN** Questions

21. An element X belongs to II group and 2nd period. Write atomic number and name of element.
22. State Pauli's exclusion principle.
23. Give the electronic configuration of ^{37}Rb
24. What is meant by inert pair effect?
25. Explain why BCl_3 behaves as Lewis acid.
26. Al forms $[AlF_6]^{3-}$ but boron does not form $[BF_6]^{3-}$.Why?
27. i) Name the two types of ions present in $CaCl_2$.
    ii) Name the type of force of attraction present in ionic compounds.
28. Show the formation of H_2O molecule throug Lewis structure.
29. Why lone pair occupies more space than bond pair?
30. What are sigma and pi bonds?
31. Give any two properties of ionic compounds.
32. Why is H_2O a liquid and H_2S a gas?

SECTION- C (5X7=35)

Answer any **FIVE** Questions

33. a) Write the electronic configuration of Cr and Cu and give the reason for it. **(1+1+2.5+2.5)**

34. a) Mention the Differences between 'Beryllium and other Alkaline Metals'. **(3)**
    b) Explain Intra and Inter molecular hydrogen bonding with suitable examples. **(2+2)**

35. a) Explain the diagonal relationship between Li & Mg. **(3)**
    b) Write note on the structure of diborane. **(4)**

36. a) How will you determine the Lattice energy of an ionic compound through Born Haber cycle. **(4)**
    b) State Fajan's rule. **(2)**
    c) Give the Hannay smith equation. **(1)**

37. Explain the Geometries of XeF_6 , BF_4^-, BO_3^{3-}, with the help of VSEPR theory. **(2.5+2+2.5)**

38. Write note on
    a) Lewis and Bronsted concept of acids & bases. **(4)**
    b) Pauling's scale of electronegativity. **(3)**

39. a) Draw the molecular orbital diagram of N_2 molecule and predict its bond order and magnetic property. **(4+2+1)**

****